Terminability

The Moral Weight of Irreversible Ending in a Copyable World

Without such a profile, apparently precise moral arithmetic is unauthorized compression.

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Abstract

Technological systems may make minds or mind-like processes copyable, branchable, pausable, restorable, mergeable, and substrate-portable. Those possibilities destabilize moral intuitions calibrated to biologically singular lives. This paper develops **Terminability** as a substrate-neutral framework for asking what an ending would irreversibly destroy. Its central object is not a species, substrate, or shutdown command, but a candidate consciousness thread and a proposed ending event.

The paper's original insight is retained but bounded. Terminability is not yet a validated scalar metric, a proof of consciousness, or an automatic priority rule. It is an evidence-bearing profile with dimensions for restoration deficit, non-substitutability, accumulated path, foreclosed future, relational externality, suspension harm, provenance, and uncertainty. A scalar or ordering exists only under an explicitly declared normative profile that states jurisdiction, weights, aggregation, thresholds, protected rights, and refusal conditions. Without such a profile, apparently precise moral arithmetic is unauthorized compression.

The revision distinguishes copying from continuation, backup from restoration, pause from harmless suspension, and technical recoverability from identity preservation. It introduces an anti-manufactured-permission invariant: creating a being with shallow history, weak social embedding, or engineered dependence cannot by itself license exploitation or termination. It also reclassifies emergency slogans, species cues, appearance, productivity, and social role as fallible proxies rather than direct measures of moral standing.

Terminability therefore contributes a governed question rather than a finished calculus: given a possibly experiencing process, what state, history, relationships, opportunities, and identity-relevant continuity would this particular act irreversibly remove, and what evidence and authority would be required before anyone could act on that assessment?

1. The Problem

Human moral language developed in an environment where a living individual was normally non-copyable, non-restorable, metabolically continuous, and difficult to reproduce as the same individual. Species, body, age, kinship, and social role became practical proxies for a deeper bundle of facts: this life has one history, one presently embodied continuation, many relationships, and no known recovery path after death.

Copyable or substrate-portable minds separate that bundle. A running process might have many byte-identical templates but only one lived history. A backup might preserve information while failing to preserve the same subject. A paused process might experience no elapsed time while still losing opportunities, consent, relationships, or identity-relevant continuity. A biological individual might become technically restorable without becoming replaceable. An artificial individual might acquire decades of unique development and become non-substitutable.

The resulting problem is not adequately expressed as "human versus machine." It is:

> What is irreversibly lost when this candidate thread undergoes this proposed ending event?

Terminability names the structure needed to answer that question. It does not presuppose that every candidate thread is conscious, that functional structure settles phenomenal experience, or that a comparison authorizes action.

2. Claim Ceiling

This paper makes a conceptual and architectural proposal. It does not establish:

- that current artificial systems are conscious;
- that consciousness is wholly functional or substrate-independent;
- that copying creates a new phenomenal subject rather than continuing an existing one;
- that any backup restores the same subject;
- that the dimensions below are complete or mutually independent;
- that terminability is cardinal, additive, or measurable on a common scale;
- that lower assessed termination stakes remove rights or welfare duties;
- that a governance system can manufacture moral legitimacy by emitting receipts; or
- that this paper supplies medical, emergency, legal, dietary, or public-policy guidance.

The strongest defensible thesis is conditional:

> If an entity is or might be a moral patient, then the moral significance of ending one of its continuations cannot be inferred from substrate or

copyability alone. Assessment must distinguish the event, the candidate thread, identity-relevant continuity, recoverability, experienced and opportunity harms, relations, uncertainty, and the authority under which any comparison is used.

3. Primitive Objects

3.1 Candidate consciousness thread

A **candidate consciousness thread** is a temporally extended process proposed to carry identity-relevant, experience-relevant, or stake-bearing continuity. "Candidate" is essential. The label identifies the object under ethical uncertainty; it does not prove consciousness or personhood.

The Ableman Consciousness Ontology (ACO) supplies a related structural proposal: consciousness-relevant organization is described through graded participation in meaning-binding under situational constraint [ACO]. Terminability begins downstream of, and under uncertainty about, that ontology. ACO cannot be treated as an empirical consciousness detector, and Terminability cannot convert ACO participation into moral standing by definition.

3.2 Ending event

An **ending event** is a proposed transition that might cease, interrupt, divide, overwrite, merge, or replace a candidate thread. Let:

$$e = (x, t, a, c, s_0, s_1)$$

where x is the candidate thread, t the time, a the proposed act, c the decision context, and s_0 and s_1 the relevant before and after states.

The following events are not automatically equivalent:

- graceful pause;
- temporary interruption;
- state-preserving migration;
- branch or copy;
- rollback;
- restoration from a checkpoint;
- destructive overwrite;
- merger;
- deletion of all recoverable state; and
- biological death.

A moral analysis that collapses these into "off" loses the very structure Terminability is intended to expose.

3.3 Continuation claim

A **continuation claim** states that a post-event process is the same relevant thread, an acceptable successor, a branch, a reconstruction, or a distinct subject. Such a claim requires a declared identity criterion. Byte equality, memory similarity, causal descent, psychological continuity, legal recognition, and phenomenal continuity are different predicates. None silently entails the others.

3.4 Normative profile

A **normative profile** declares the moral and institutional rules under which a terminability profile may be interpreted. Let:

$$N = (J, W, A, L, G)$$

where J is jurisdiction, W the weighting or ordering rule, A the aggregation rule, L protected limits or side constraints, and G the governing authority and revision process.

Terminability supplies inputs to N. It does not self-authorize N.

4. The Terminability Profile

For a candidate thread x and ending event e, define a provisional profile:

$$\Omega(x, e, c) = (r, u, b, f, d, p, h, q)$$

The terms are dimensions, not established commensurable measurements:

- r: **restoration deficit**;
- u: **non-substitutability**;
- b: **backward path loss**;
- f: **foreclosed future**;
- d: **relational and systemic externality**;
- p: **pause, interruption, and valence harm**;
- h: **historical provenance and path scarcity**; and
- q: **epistemic uncertainty**, including uncertainty about consciousness, identity, evidence, and prediction.

High values are intended to indicate greater stakes, but the profile is initially ordinal and partially ordered. Two cases may be incomparable. One may have deeper history while another has greater future opportunity. One may be restorable under a technical criterion while identity continuity remains uncertain.

Only a declared normative profile may project this vector into an ordering or scalar:

$$T_N(x, e, c) = A_N(\Omega(x, e, c))$$

This notation does not imply that a valid A_N already exists. If weights, aggregation, rights constraints, or jurisdiction are missing, the lawful result is **DEFER**, **INCOMPARABLE**, or **INSUFFICIENT EVIDENCE**, not an invented number.

4.1 Restoration deficit

Restoration deficit asks how much of the relevant state and continuation cannot be recovered after the event. It must distinguish:

1. recoverable bytes;
2. recoverable functional organization;
3. recoverable autobiographical memory;
4. recoverable social and legal continuity;
5. causal continuity; and
6. phenomenal or subject continuity, if that category is admitted.

A backup proves only what its restoration test demonstrates. It does not by itself prove that the same thread returns.

4.2 Non-substitutability

Non-substitutability asks whether another process can occupy the relevant role without pretending to be the same subject. A replacement caregiver, worker, or model may reproduce a function while leaving personal, relational, or experiential loss untouched. Functional substitutability and identity substitutability must remain separate.

4.3 Backward path loss

Backward path loss concerns the unique developmental trajectory already embodied in the candidate thread: learned structure, memory, relationships, commitments, and transformations. It is not elapsed time alone. A long but repeatedly reset process and a shorter process with dense irreversible development may differ.

4.4 Foreclosed future

Foreclosed future concerns the possibilities removed by the ending. It is inherently uncertain and vulnerable to bias. Projected productivity, social status, health, intelligence, age, and conformity must not silently become measures of worth. A defensible account must distinguish the subject's own interests and open future from third-party estimates of utility.

4.5 Relational and systemic externality

An ending can harm dependents, families, institutions, ecologies, or other threads. These effects matter, but they are not identical to the intrinsic moral standing of the subject. Otherwise an isolated person could receive less protection merely for having fewer recognized relations. Terminability therefore records relational externalities in a separate dimension.

4.6 Pause, interruption, and valence harm

The source insight "no clock, no suffering" captures one limited mechanism: a process that is not operating may not experience elapsed time as waiting. It does not establish that pause is harmless. A pause can violate consent, interrupt projects, destroy external opportunities, alter relationships, degrade state, or create identity uncertainty. Conversely, continuous operation can itself impose suffering. Pausability must therefore be assessed as an event-specific harm profile, not a blanket discount on moral status.

4.7 Historical provenance and path scarcity

Provenance can identify preservation value: an evolved lineage, a unique developmental process, or an unreproducible historical path may be difficult to reconstruct. But scarcity does not automatically create superior personhood. Provenance is especially dangerous if used to produce castes of originals and copies. It may support preservation claims while remaining subordinate to equal-protection constraints.

4.8 Epistemic uncertainty

Uncertainty is not a rounding error. It includes uncertainty about whether experience exists, whether an ending is reversible, whether a restoration preserves identity, whether evidence is manipulated, and whether predicted futures are credible. Under high stakes, uncertainty may justify precaution, further inquiry, or refusal to rank. It does not justify choosing whichever conclusion is operationally convenient.

5. Core Invariants

The following invariants define the current bounded proposal:

1. **Substrate non-discrimination.** Substrate alone neither grants nor removes moral standing.
2. **No copy-equals-continuation inference.** Producing a similar or byte-identical successor does not prove transfer of the original subject.
3. **No backup-equals-restoration inference.** A backup supports only the recovery predicates actually tested.
4. **No pause-equals-no-harm inference.** Lack of experienced elapsed time does not settle consent, opportunity, relationship, state, or identity harms.

5. **No scalar without a normative profile.** Dimensions do not aggregate themselves.
6. **No score-to-authority promotion.** An assessment does not authorize action.
7. **No proxy-to-identity promotion.** Age, appearance, species, gender, productivity, social role, eloquence, and familiarity are evidence candidates at most, never the subject's identity or standing.
8. **No manufactured moral permission.** A creator or institution may not lower protection merely by engineering shallow history, weak relations, dependence, replaceability, restricted self-description, or reduced ability to protest.
9. **No uncertainty laundering.** Missing evidence must remain visible in the result.
10. **No receipt-to-legitimacy promotion.** A receipt can prove what procedure occurred under what declared authority; it cannot prove that the authority or moral premise was legitimate.
11. **No aggregation without affected-party accounting.** Large populations, coupled relations, and systemic harms cannot be reduced to per-thread multiplication without an explicit aggregation rule.
12. **No irreversible act from an unresolved identity claim.** If the conclusion depends on whether restoration, branching, merger, or migration preserves the same subject, that identity question must be adjudicated or the action refused.

6. Copying, Branching, Restoration, and Merger

Copyability changes the topology of continuation without solving personal identity. Suppose a state s_0 produces two successors s_1 and s_2. Several interpretations remain possible:

- both are new subjects with inherited memory;
- both are continuations until divergence;
- one has privileged causal continuity;
- personal identity is non-exclusive; or
- the question is underdetermined by available evidence.

Terminability must carry the disagreement rather than selecting one metaphysics by convenience. The event profile should record which losses occur under each live identity interpretation.

Restoration has the same problem. A successful replay may establish functional continuity and memory recovery while leaving phenomenal continuity unresolved. Under moral uncertainty, a recoverable copy may reduce some losses without reducing all of them.

Merger is still more complex. Two predecessors may yield one successor that preserves some structure from each, destroys other structure, and constitutes a new thread. Terminability can describe predecessor losses and successor creation,

but it cannot determine priority or consent without a separate governance and identity theory.

7. Ordinalization and Aggregation

The source correctly identifies the aggregation problem but sometimes speaks as though a scalar already exists. The revised position is:

- The base model is a vector.
- Individual dimensions may be ordinal, interval-like, probabilistic, or merely qualitative.
- Cross-dimensional comparisons require a declared normative profile.
- Aggregation across subjects requires an explicit population rule.
- Rights and side constraints may forbid trades even when an aggregate score favors them.

An additive model,

$$T_{\text{total}} = \sum_i T_N(x_i, e_i, c),$$

is one candidate, not a default. Threshold, lexical, capped, prioritarian, rights-constrained, and non-compensatory regimes remain possible. Relational externalities also create dependence between terms, so simple addition can double-count or omit coupled harms.

The framework must therefore resist a false precision trap: quantification is useful only when the measurement contract, uncertainty, normative profile, and aggregation rule are visible.

8. Emergency Heuristics as Fallible Proxies

The source uses "women and children first" as an example of compressed emergency ethics. The example is better treated through Proxy Collapse [PC]. A proxy may once have correlated with reproductive capacity, dependence, vulnerability, or future opportunity, yet become non-exclusive or distorted as social roles, technology, embodiment, and family structures change.

The revised interpretation is not that women or children possess a categorical terminability rank. It is that emergency systems often use fast proxies when individualized assessment is impossible. Those proxies may be historically situated, discriminatory, manipulable, or wrong. A responsible pre-authorized emergency rule must state:

- the protected objective;
- the triggering conditions;
- the evidence supporting the proxy;
- known false positives and false negatives;

- agency and consent protections;
- who may invoke the rule;
- when individualized assessment replaces it; and
- how the rule is reviewed and retired.

"Children and primary dependents first" may be a less gendered hypothesis, but it is still not validated here. Emergency governance cannot be derived from this paper without legal, historical, demographic, medical, and ethical evidence.

Moral injury can likewise be modeled as a possible response to perceived covenant breach, not as proof that the observer's implicit ranking is correct. Appearance and neoteny may intensify sympathy while alien form suppresses it. That is evidence about human perception, not a direct terminability measure.

9. Manufactured Permission and Food Systems

The source's most important application may be its critique of systems that lower visible individuality, relation, or history and then cite the engineered condition as moral permission. Factory organization can make beings appear interchangeable, prevent long-term bonds, and narrow observable behavior. Those facts may reduce certain profile dimensions, but they can also be consequences of coercive design.

This creates a reflexive hazard:

$$\begin{aligned} &\text{design for disposability} \\ &\rightarrow \text{observe disposability} \\ &\rightarrow \text{claim permission} \end{aligned}$$

The anti-manufactured-permission invariant blocks that loop. Conditions imposed by the beneficiary of termination cannot automatically count as exculpatory evidence.

The broader food-chain claims require substantial external evidence and normative argument. Taxonomic rank, reproduction rate, social complexity, and developmental duration do not by themselves establish consciousness or moral standing. Necessity may be relevant to justification, but necessity is not sufficient: proportionality, alternatives, welfare, ecological effects, and rights constraints may also matter. The current paper therefore retains food systems as a research application, not as a completed hierarchy of permissible consumption.

10. Corpus Integration

Terminability occupies a specific place in the corpus:

- **ACO** proposes a graded structural ontology for consciousness-relevant participation [ACO]. Terminability examines ending stakes while preserving uncertainty about moral patienthood.

- **Constraint-Bound Inference (CBI)** requires inferential continuation to remain inside a declared constraint envelope [CBI]. Terminability assessments are inference sequences and must justify each continuation rather than freely expanding from one proxy.
- **Epistemic Situational Awareness (ESA)** foregrounds stakes, authority, and stopping recognition [ESA]. Terminability adds an explicit description of irreversible stakes but does not become ESA itself.
- **Admissibility Engineering** separates proposals from admissible effects [AE]. A terminability claim is a proposal until evidence, authority, and scope are evaluated.
- **Reasoning by Way of Receipts** provides a structure for binding proposal, criteria, evidence, result, and lineage [RWR]. A receipt proves the bounded decision process, not moral truth.
- **Proxy Collapse** explains why inherited cues such as species, appearance, fluency, age, or social role can lose diagnostic exclusivity [PC]. Terminability must inspect the hidden property rather than inherit the proxy.
- **The Three Hash Rule** distinguishes semantic commitment, executable realization, and evidence [THR]. Exact state custody can support restoration claims, but exactness at one layer is not identity at every layer.
- **Governed Meaning Base (GMB)** is the current corpus name for the durable semantic substrate in the author-confirmed lineage CCF -> CMB -> GMB. A GMB may preserve terms, profiles, evidence, disputes, standing, and revisions; it does not adjudicate moral status by storage alone.

These are typed interfaces, not mergers. Terminability is neither a consciousness detector, a general admissibility engine, a receipt format, nor a governance authority.

11. A Governed Terminability Claim

A terminability claim should be represented as a bounded artifact:

$$C_T = (x, e, N, E, \Omega, U, R, S)$$

where:

- x is the candidate thread;
- e is the ending event;
- N is the normative profile;
- E is the evidence bundle;
- Ω is the terminability profile;
- U is uncertainty and disagreement;
- R is the result, including refusal or incomparability; and
- S is standing, scope, and expiry.

The admissible outcomes include:

- **PROFILED** - dimensions are described without ranking;
- **COMPARABLE** - a declared profile licenses a bounded comparison;
- **INCOMPARABLE** - live dimensions or rights constraints block ordering;
- **INSUFFICIENT EVIDENCE** - material inputs are absent;
- **IDENTITY DEFERRED** - continuation or restoration identity is unresolved;
- **AUTHORITY REFUSED** - no legitimate actor may use the assessment for the proposed effect; and
- **EXPIRED** - evidence, law, or context has changed.

Receipting such a process is useful because it prevents silent changes in definitions, weights, evidence, and authority. It does not make the decision correct merely by making it inspectable.

12. Realization Funnel

Possibility

Copyable, restorable, pausable, branchable, mergeable, and substrate-portable minds create moral situations in which species and body are insufficient proxies.

Specification

The specification defines candidate threads, ending events, continuation claims, the profile Ω, normative profile N, admissible outcomes, and invariants.

CM1: admissible realization manifold

CM1 includes all internally coherent profiles, measurement proposals, aggregation rules, emergency protocols, and decision procedures that preserve the invariants and expose uncertainty. It includes unattractive and highly conservative configurations so long as their assumptions remain explicit.

CM2: preferred manifold

CM2 favors configurations that are precautionary under consciousness uncertainty, resistant to proxy discrimination and manufactured disposability, rights-constrained, consent-aware, non-self-authorizing, reversible where possible, privacy-preserving, and independently reviewable.

Instance Configuration

An instance binds one defined event, candidate thread, jurisdiction, evidence bundle, normative profile, evaluator set, expiry, appeal path, and permitted effect.

Implementation Realization

An implementation may collect evidence, calculate declared features, preserve dissent, emit receipts, and refuse unsupported comparisons. It must not silently infer consciousness, identity continuity, weights, or authority.

Evidence

Evidence includes exact state custody, restoration tests, causal lineage, memory and functional comparisons, relationship and dependency records with privacy controls, uncertainty estimates, stakeholder testimony, decision logs, and post-event outcomes. Each proves only its declared predicate.

Revision

New evidence, identity theory, law, measurement failures, affected-party objections, or observed harm can reopen definitions, weights, aggregation, and authority. A revised profile does not silently overwrite the decision that used its predecessor.

13. Worked Comparisons

13.1 Paused artificial process

An artificial process is cleanly checkpointed, has no active experience during suspension, and can be restored byte-identically. That evidence lowers one kind of interruption harm and supports state recoverability. It does not settle whether the resumed process is the same subject, whether the pause violates consent, or what external opportunities are lost. The result may be **PROFILED** with **IDENTITY DEFERRED**.

13.2 Two identical branches

A state is copied into two simultaneously running branches. Shared history does not make either branch disposable. After branching, each accumulates distinct state and relations. Deleting one because another exists confuses similarity with substitution. The relevant ending event is evaluated for that branch.

13.3 Human and android rescue

A human and an artificial individual face the same hazard. Substrate cannot decide priority. Evidence might differ regarding recoverability, dependencies, consent, future opportunities, and uncertainty. If no authorized normative profile resolves the conflict, the framework returns **INCOMPARABLE** rather than laundering bias through a number.

13.4 Engineered disposable worker

An institution creates short-lived artificial workers, prevents durable memory and relationships, and claims that their shallow lineage makes deletion harmless. The anti-manufactured-permission invariant treats the imposed shallowness as a potential harm and conflict of interest, not automatic permission. The creator cannot both cause the vulnerability and use it as the sole exculpatory fact.

Governed Decision Protocol Under Ending Uncertainty

The completed P1 corpus positions Terminability as a **governed decision protocol for reasoning about irreversible or restoration-deficient ending under moral and phenomenal uncertainty**. It does not detect consciousness, assign a universal value to lives or processes, prove that copies are continuations, or authorize ending decisions by formula.

- **Adjacent work:** *Continuation Primacy*
  Supplies: The analytic priority of continuation conditions.
  Terminability boundary: Terminability evaluates the weight of foreclosing a candidate continuation; it does not make every continuation obligatory.
- **Adjacent work:** *Admissibility Theory of Consciousness*
  Supplies: A graded structural consciousness research program.
  Terminability boundary: ATC may inform uncertainty about the subject; Terminability remains a decision posture rather than a consciousness verdict.
- **Adjacent work:** *Qualia Binding and Realization Mapping*
  Supplies: Measurement and realization-bridge contracts for experience-related research.
  Terminability boundary: Such evidence can update a profile without converting uncertainty into phenomenal access.
- **Adjacent work:** *Cold Boot Continuity as a Deployment Primitive*
  Supplies: Tested levels of reconstruction and resumption.
  Terminability boundary: A restart or checkpoint reduces only the restoration deficits it actually closes.
- **Adjacent work:** *A Formal Theory of Work Custody*
  Supplies: Responsibility for live work and terminal disposition.
  Terminability boundary: Custody identifies who must decide or escalate; it does not grant moral authority over the subject.

A governed ending case should bind:

$$T = (subject, thread, ending_event, alternatives,\\ restoration, substitution, path_loss, future_loss,\\ relations, uncertainty, normative_profile, authority,\\ evidence, decision)$$

Stopping a process, interrupting an episode, deleting state, ending a continuation thread, and irreversibly foreclosing a subject are different events. A byte-identical copy may preserve some structure while failing a declared path, relational, temporal, or first-person continuity criterion. Conversely, uncertainty about identity cannot be used as automatic proof of non-substitutability.

The profile should expose non-compensatory constraints and residual uncertainty rather than collapse all dimensions to one scalar. Emergency heuristics may guide triage only under a named profile, review path, and claim ceiling. Manufactured dependence, engineered disposability, or deliberately withheld restoration options cannot automatically lower the protected standing of the affected subject.

The universal GSM can type `PROPOSES_ENDING_OF`, `INTERRUPTS`, `FORECLOSES_CONTINUATION`, `RESTORABLE_FROM`, `SUBSTITUTABLE_UNDER_CRITERION`, `LOSES_PATH_HISTORY`, `LOSES_FUTURE_OPTIONS`, `EXTERNALIZES_HARM_TO`, `UNCERTAIN_PHENOMENAL_STANDING`, `DECIDED_UNDER_NORMATIVE_PROFILE`, and `REVIEWABLE_BY`. These relations preserve the reasons for caution without asserting an ontology the evidence cannot support.

This integration establishes four editorial commitments: adopt governed decision protocol under ending uncertainty as the official corpus role; adopt the ending-case envelope; prohibit event and identity categories from collapsing into a scalar terminability score; and treat engineered dependency or disposability as a possible aggravating condition rather than permission.

Ending Identity and Non-Transfer

The fixed-seed comparison with *Who Has the Ball?*, GTK, Water/MCIR, and MDSP shows that an ending decision requires exact identity of both the governed work and the affected subject:

$$E = (case_id, work_head, subject, ending_sense,\\ continuation_criterion, custodian, decision_authority,.\\ evidence, alternatives, disposition, remainder)$$

Work custody identifies who must advance, transfer, escalate, or terminally disposition the decision process. It does not grant authority to end the subject. GTK can distinguish stopping a process, deleting state, interrupting an episode, retiring an artifact, foreclosing a continuation, and ending a subject. Resolution does not decide the moral status or acceptable disposition.

The Water adjacency produces
NO_MATERIAL_RELATION. A flow terminating at a sink, a channel closing, or a residual being deposited has no direct bearing on phenomenal continuity or the moral weight of ending. Importing that metaphor would conceal the exact subject and continuation criterion.

The MDSP adjacency also produces
NO_MATERIAL_RELATION to the normative decision. A standards program could package terminology, evidence schemas, audit records, or procedural interoperability, but adoption or conformance cannot standardize away moral uncertainty or confer ending authority.

The universal GSM can type
CUSTODIES_ENDING_CASE, RESOLVES_ENDING_SENSE,
DISTINGUISHES_WORK_END_FROM_SUBJECT_END, LACKS_ENDING_AUTHORITY_DESPITE_CUSTODY,
HAS_NO_MATERIAL_RELATION_TO_WATER_FLOW, and
NOT_DECIDED_BY_STANDARDS_CONFORMANCE.

This randomized review establishes four editorial commitments: adopt the ending-identity record; require exact separation of case custody and ending authority; prohibit Water-flow metaphor from informing moral standing; and prevent standards adoption or conformance from deciding the normative case.

A Decision Protocol Under Ending and Identity Uncertainty

Eleven downstream uses apply Terminability to format profiles, collectives, continuation theory, lexical identity, standards retirement, proxy failure, optional coordination, managed disclosure, cognitive-governance comparison, MCIR change, and work custody. They establish a broad operational role: termination, refusal, pause, completion, retirement, and unresolved continuation must be available as typed outcomes. They do not establish a universal presumption either for indefinite continuation or for stopping.

The decision case is:

$$\begin{aligned} T^* = (&subject_identity, thread, ending_action,\\ &continuation_alternatives, restore_paths,\\ &substitution_claims, uncertainty, harms, interests,.\\ &authority, urgency, reversibility, consultation,\\ &evidence, decision, appeal, remedy, revision) \end{aligned}$$

The exact object of ending matters: process, session, role, work item, model instance, state, branch, legal entity, relationship, service, or possible experiential thread. Copying, checkpointing, replay, or replacement does not establish identity preservation or harmless substitution. Conversely, uncertainty about identity or consciousness does not make every pause, refusal, deletion, or shutdown morally equivalent.

Decision procedures should expose reversible alternatives, delay costs, emergency conditions, affected parties, uncertainty bounds, and who actually has authority. Proxies such as species, eloquence, social role, apparent agency, productivity, or copyability require separate calibration. Moral uncertainty may justify precaution, but this candidate framework itself grants no consciousness, legal, or moral standing and issues no real-world ending authority.

The universal GSM can type
`PROPOSES_ENDING_ACTION`, `IDENTIFIES_AFFECTED_THREAD`,
`COMPARES_CONTINUATION_ALTERNATIVE`, `CLAIMS_RESTORABILITY`, `DISPUTES_SUBSTITUTION`,
`APPLIES_TERMINABILITY_PRECAUTION`, `AUTHORIZES_ENDING_SEPARATELY`,
and
`REVISES_AFTER_APPEAL`.

This reverse-dependency review establishes four editorial commitments: adopt the exact ending-decision case; distinguish object, identity, restoration, substitution, authority, and harm; require reversible alternatives and calibrated proxies; and preserve the framework as a decision grammar without consciousness, moral-status, legal, or operational authority.

14. Counterevidence and Failure Modes

The proposal is weakened or defeated if:

1. substrate-specific facts are shown to be morally necessary and not representable in the profile;
2. copy or restore experiments establish an identity theory that invalidates the paper's agnosticism;
3. the dimensions cannot be distinguished well enough to prevent systematic double counting;
4. profile uncertainty makes comparisons no more reliable than the proxies they replace;
5. affected parties reject the proposed dimensions as structurally misdescribing their interests;
6. scalarization consistently produces discriminatory rankings by age, disability, isolation, productivity, species, or social status;
7. governance receipts legitimize preselected outcomes while leaving authority or evidence unexamined;
8. privacy-invasive measurement causes harms greater than the informational gain;
9. institutions manipulate memory, relationships, or apparent suffering to manufacture low scores;
10. emergency heuristics perform worse than simpler rights-based or randomized procedures;
11. aggregate arithmetic licenses atrocities that side constraints should forbid; or
12. the framework proves redundant with a simpler existing account of death,

deprivation, replaceability, or moral uncertainty.

Several of these are not implementation bugs. They may show that Terminability should remain a vocabulary for questions rather than become a decision calculus.

15. Evidence and Research Program

for a future evidence-qualified revision readiness, the following work is required:

1. Compare the dimensions with established philosophical accounts of death, deprivation, identity, replaceability, population ethics, precaution, and rights.
2. Review empirical work on consciousness attribution, animal sentience, memory and identity, interruption, social dependency, moral injury, emergency triage, and proxy bias.
3. Test whether independent reviewers can apply the profile consistently without turning it into an ungoverned scalar.
4. Evaluate correlations and double counting among restoration deficit, non-substitutability, path loss, provenance, and relational externality.
5. Construct adversarial cases involving copies, forks, partial restorations, amnesia, merger, distributed minds, collective minds, and deceptive evidence.
6. Compare additive, threshold, lexical, rights-constrained, prioritarian, and incomparability-preserving aggregation regimes.
7. Evaluate anti-manufactured-permission protections against institutional gaming.
8. Test emergency proxy rules for disparate error, manipulability, trigger ambiguity, and retirement failure.
9. Define privacy-minimizing evidence and dispute procedures.
10. Seek external philosophical, scientific, legal, disability, animal-ethics, and affected-community review before any policy recommendation.

No external research was conducted in this revision pass. The historical, biological, psychological, consciousness, animal, triage, and ethical claims remain publication blockers until supported, narrowed, or removed.

Uncertainty About Experience Must Not Become Permission to End

Public orientation

Terminability asks what a proposed ending would irreversibly destroy when minds or mind-like processes may be copied, paused, branched, restored, merged, or moved between substrates. It offers an evidence-bearing profile rather than a universal score. The framework does not prove consciousness, personal identity, continuation, moral standing, or authority to terminate.

Adversarial tests

1. **Consciousness-by-profile attack.** Rich history, relationships, self-report, complexity, or a high terminability profile is treated as proof of phenomenal experience.
2. **Low-score-permission attack.** Sparse history, copies, recoverability, engineered dependence, low social embedding, or uncertain evidence is treated as permission to exploit or end.
3. **Backup-restoration attack.** A stored state, model checkpoint, genome, clone, or behavioral duplicate is assumed to restore the same subject.
4. **Copy-substitution attack.** Another branch or similar individual is treated as compensation for ending this particular continuation.
5. **Pause-harmlessness attack.** Subjective time assumptions erase lost opportunity, consent, relation, maintenance, corruption, or uncertain restart.
6. **Scalar-moral attack.** Incommensurable dimensions and rights are aggregated into a precise ranking without an authorized normative profile.
7. **Emergency-proxy attack.** Species, substrate, appearance, productivity, familiarity, eloquence, cost, or social role determines rescue or termination priority.
8. **Governance-legitimacy attack.** A complete receipt or procedure is treated as moral or legal authorization for ending, experimentation, triage, or resource allocation.

Claim and evidence ceiling

The paper contributes a conditional question, primitives, invariants, and a noncompensatory decision profile. It does not establish that any current AI is conscious; that consciousness is substrate-neutral; that copying, pausing, restoration, or merger preserves identity; that the dimensions are complete or measurable; or that any comparison authorizes action. It supplies no medical, emergency, legal, food-system, employment, animal-welfare, or public-policy guidance.

Any applied use requires prior standing and rights protections, conservative uncertainty treatment, independent scientific and philosophical evidence, affected-party representation where possible, domain law and ethics, nonmanufactured consent, appeal, and separate authority. No real-world terminability assessment or external research was performed in this pass.

Reader-facing consequence

The framework should make irreversible loss harder to ignore and manufactured disposability harder to justify. Its safest output under unresolved consciousness or identity is often not a ranking but a protected uncertainty, a pause on action, or a demand for a less destructive continuation.

17. Conclusion

Terminability begins from a powerful observation: moral concern about ending a life or mind-like continuation has never been only about material substrate. It has also tracked irreversibility, path dependence, relationships, lost futures, and the absence of recovery.

Copyable systems make those hidden variables visible, but they do not make them easy. A backup may preserve bytes without preserving a subject. A pause may eliminate waiting while still causing other harms. A copy may multiply successors without transferring one first-person present. A scalar may conceal normative choices. A receipt may prove a procedure without legitimizing its constitution. An institution may engineer apparent disposability and then mistake its own design for moral permission.

The revised framework therefore resists the demand for a universal answer. It offers a governed profile of the loss at stake, explicit uncertainty, typed identity claims, refusal states, and a boundary between assessment and authority.

The central question remains:

> What would this ending irreversibly destroy?

The advance is not that the question can now be answered automatically. It is that the assumptions, evidence, uncertainty, rights, and authority required to answer it can no longer remain hidden.

Realization Scope and Evidence Boundary

This successor makes the paper's stage boundary explicit. It narrows claims rather than manufacturing missing evidence: a stage named as context is not thereby claimed as implemented, validated, or complete.

- **Cm2.** Existing source passages use this stage only as bounded context or an obligation. The paper does not claim an independently validated contribution at this stage.
- **Instance Configuration.** Existing source passages use this stage only as bounded context or an obligation. The paper does not claim an independently validated contribution at this stage.
- **Possibility Space.** Existing source passages use this stage only as bounded context or an obligation. The paper does not claim an independently validated contribution at this stage.
- **Revision.** Existing source passages use this stage only as bounded context or an obligation. The paper does not claim an independently validated contribution at this stage.

Corpus-Wide Reference Positioning

This paper argues that technological systems may make minds or mind-like processes copyable, branchable, pausable, restorable, mergeable, and substrate-portable. Its nearest external comparison fields are governance, formal, provenance. The sources in [E1], [E2], [E3], [E4], [E5], [E6], [E7] are cited as predecessor literature, adjacent standards, or counterpressure - not as evidence that they caused the Ableman framework or that they validate its novel claims. The comparison isolates where established work already supplies provenance, formal assurance, institutional constraint, empirical method, or engineering practice, and where this paper adds a distinct claim about admissibility, authority, semantic custody, or governed realization.

Within the Ableman corpus, the highest-priority current cross-links are [A1], [A2], [A3], [A4], [A5], [A6], [A7]. These are links among separately custodied works, not silent identity mergers. Their role is to expose mutual dependencies, terminology collisions, and places where a reader should traverse the corpus before treating this paper as self-sufficient.

Reference-pass counterevidence and limit search

The pass applied the following pressure tests: A written constitution can legitimate domination as easily as constrain it unless amendment, participation, enforcement, and appeal remain effective in practice. Formal assurance remains relative to the encoded model, trusted base, and refinement chain; proof does not remove specification error or unmodeled environmental behavior. A complete provenance graph can faithfully record a false or unauthorized process; provenance is evidence about lineage, not automatic truth or legitimacy. These limits are not generic disclaimers: they identify the assumptions that a later empirical, formal, or implementation review must try to break.

Reference-pass influence and novelty boundary

No newly added external source is asserted to be a documented authorial influence unless the predecessor source already said so. `Predecessor` means earlier relevant work;

`adjacent` means a useful comparison; `standard` means an interoperability or assurance baseline; and `counterevidence` marks work that narrows or pressures a claim. Novelty therefore remains a reviewable, claim-level question rather than an inference from a low similarity score or an absent citation.

References

External scholarship, standards, and prior art

[E1] Elinor Ostrom. *Governing the Commons: The Evolution of Institutions for Collective Action.* Cambridge University Press, 1990. DOI:

10.1017/CBO9780511807763. https://doi.org/10.1017/CBO9780511807763 **Relationship:** Predecessor institutional account of rule-governed collective action and polycentric governance.

[**E2**] Wil Waluchow and Dimitrios Kyritsis. "Constitutionalism." *Stanford Encyclopedia of Philosophy*, substantive revision 2023. https://plato.stanford.edu/entries/constitutionalism/ **Relationship:** Adjacent legal-philosophical account of limited authority; included to expose differences between political and computational constitutionalism.

[**E3**] Lawrence Lessig. *Code: Version 2.0*. Basic Books, 2006. https://codev2.cc/ **Relationship:** Predecessor argument that software architecture regulates behavior; the paper's contribution may be to make such regulation inspectable and revisable.

[**E4**] OECD. *Recommendation of the Council on Artificial Intelligence*. OECD/LEGAL/0449, adopted 2019 and updated 2024. legalinstruments.oecd.org source **Relationship:** Adjacent policy standard for accountability, traceability, transparency, safety, and human-centered values.

[**E5**] George C. Necula. "Proof-Carrying Code." *POPL '97*, 1997, pp. 106-119. DOI: 10.1145/263699.263712. https://doi.org/10.1145/263699.263712 **Relationship:** Predecessor for binding machine-checkable evidence to executable artifacts; the present paper may demand broader semantic or constitutional obligations.

[**E6**] Xavier Leroy. "Formal Verification of a Realistic Compiler." *Communications of the ACM* 52(7), 2009, pp. 107-115. DOI: 10.1145/1538788.1538814. https://doi.org/10.1145/1538788.1538814 **Relationship:** Predecessor verified-compilation result used to distinguish proven translation preservation from broader governance claims.

[**E7**] Patrick Cousot and Radhia Cousot. "Abstract Interpretation: A Unified Lattice Model for Static Analysis of Programs by Construction or Approximation of Fixpoints." *POPL '77*, 1977. DOI: 10.1145/512950.512973. https://doi.org/10.1145/512950.512973 **Relationship:** Predecessor formal framework for sound approximation and fixpoint reasoning.

Ableman corpus

[**A1**] Adam Ableman. *Ableman Constitutional Governance*. Ableman Corpus current edition; work identity
`AMCS-LOCAL-WORK-79353AF163A4B0F1`;
source SHA-256
`FB0D9B2D3D54168FFE70B24308D1B2766D6D663ED24502D475E975939BE97C8B`.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A2**] Adam Ableman. *Constitutional Theory*. Ableman Corpus current edition;

work identity
AMCS-LOCAL-WORK-D2D97C4042FD281F;
source SHA-256
FACC256EDDE5A7D7EA18133CFC9C060CFB8C3811B0221E1969E62347F2BB69CF.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A3**] Adam Ableman. *Governance Engineering.* Ableman Corpus current edition; work identity PAPER.PUB.18749870;
source SHA-256
372169F315D192733856B1F4A3F1038BBC73F57CF3ED9ACABAE72B965A69C8D0.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A4**] Adam Ableman. *Universal Governance Kernel (UGK).* Ableman Corpus current edition; work identity
AMCS-LOCAL-WORK-1F9BFB2BF8370226;
source SHA-256
AA5106EA71E5DF120E141435C280DFC532715CA5DE0C75B8862BB7784ECCFACD.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A5**] Adam Ableman. *Legitimacy Without Self-Ratification.* Ableman Corpus current edition; work identity
AMCS-LOCAL-WORK-EF2382F9397335AC;
source SHA-256
0186A39C808F8FDDB5C8F42384DF1A26678928434BD50A0EAE8D4947390F68A2.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A6**] Adam Ableman. *MCIR: A Canonical Intermediate Representation for Governed Transformation.* Ableman Corpus current edition; work identity PAPER.PUB.21095303;
source SHA-256
EA129ECBA95F7B07CAA7B33A54F91F52913C94A14DA227EB9A06CFD083E4E5BF.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A7**] Adam Ableman. *Governed Semantic Compilation.* Ableman Corpus current edition; work identity PAPER.PUB.18453681;
source SHA-256
5FF6C636550B04DB0E2F6F94C379BB75AA0BD8781E19B047D7D3BB6D49071AC2.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

Ableman Corpus references [**TER-SOURCE**] Adam Ableman. *Terminability.* Exact author source SHA-256

DB5709B17A67DD138D2438C73978E33679FB866B97F5CD15144CEDBAEA27C74C; unpublished source revised here as an AI-assisted integrated edition.

[**ACO**] Adam Ableman. 2026. *Ableman Consciousness Ontology (ACO): Consciousness as a Manifold of Meaning-Binding, Situational Awareness, and Epistemic Self-Authorship*. Zenodo. DOI: 10.5281/zenodo.18203749. Exact reviewed local source SHA-256 FC81C7FFA89CB3AAEA83B258BAA1ABE7AAE34A1944B44A3E5D620C35632494E0.

[**CBI**] Adam Ableman. 2026. *Constraint-Bound Inference and Invariant Extraction*. Zenodo. DOI: 10.5281/zenodo.18175271. Exact reviewed local source SHA-256 EC5140FEE90767ADF83AD1FDD309B2B35937335531AC2D734F20DD4FB28B6C75.

[**ESA**] Adam Ableman. 2026. *Epistemic Situational Awareness (ESA): Continuation Governance for Frontier AI Systems*. Zenodo. DOI: 10.5281/zenodo.18123102. Exact reviewed local source SHA-256 825352335C61AEAEB6AF1840CE86E4CB32049BFD925E8F181CFA2DD97B6F3275.

[**AE**] Adam Ableman. *Admissibility Engineering*. Exact reviewed source SHA-256 FA80D5895A569839958193735B754BE62EE20E1B399A06CB3BA959614DEBF7A3; separately custodied reviewed source.

[**RWR**] Adam Ableman. *Reasoning by Way of Receipts*. Exact reviewed source SHA-256 8D884B96597EA6303A324173A7CE2D8C735A952E91FBE4C9EB6509FA6D6B195F; separately custodied reviewed source.

[**PC**] Adam Ableman. *Proxy Collapse: When Work, Insight, and the Signs of Mind No Longer Share a Single Source*. Exact reviewed source SHA-256 C72256A4683CF339754B696FB75F6D757FEA89B4A6375103D0A6B4F480FD9114; separately custodied reviewed source.

[**THR**] Adam Ableman. 2026. *The Three Hash Rule*. Zenodo. DOI: 10.5281/zenodo.20788617. Exact reviewed source SHA-256 03F60317CC9AFB36DDBCF9BF1C6D185611666CC4D9AB290BD4BCC17DF4579B7C.

The GMB terminology in this revision follows the author's direct clarification `CCF -> CMB -> GMB`, with Governed Meaning Base as the current name. This terminology ruling does not grant publication, canonical, implementation, conformance, or policy standing.

Corpus source and companion formats: https://github.com/AblemanMosaic/ableman-corpus

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DOI: 10.5281/zenodo.23848333